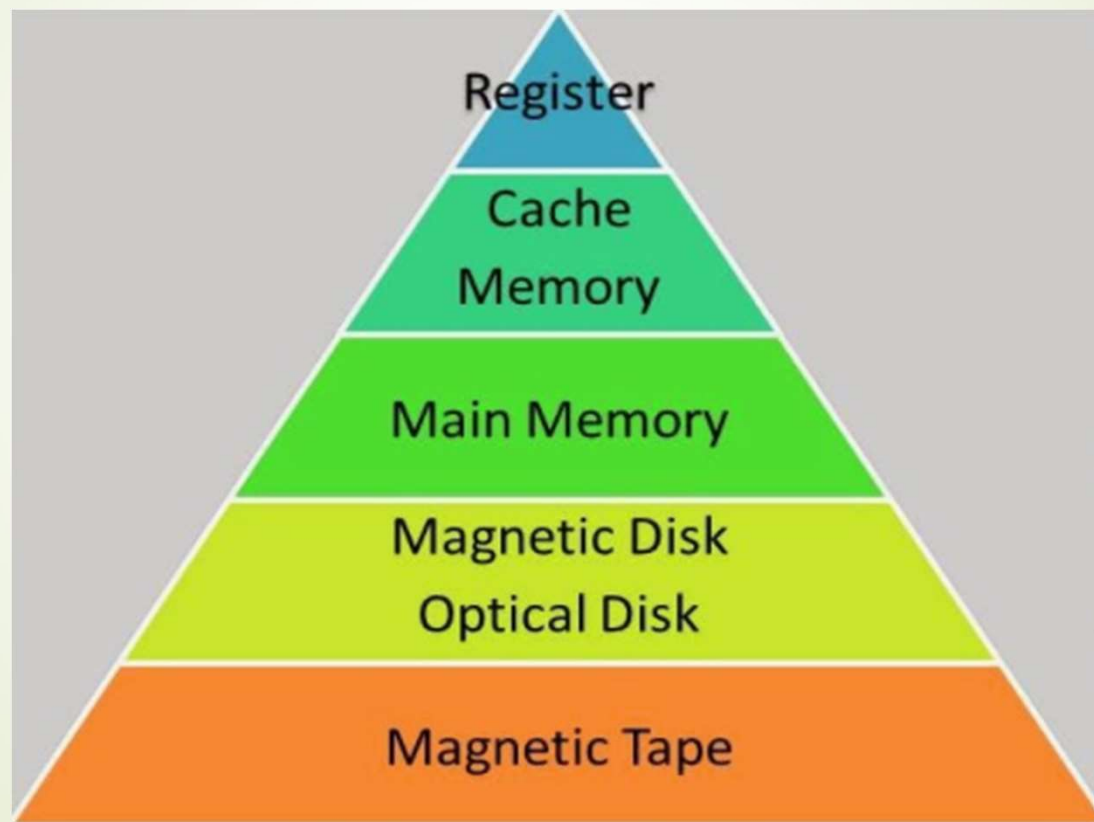


# Hierarchical memory system





## Contd...

- The design constraints on computer memory depends on:
  - capacity
  - speed / access time
  - cost
- However, there is a trade-off among these three characteristics and the following relationship is observed.
  - => lesser access time, greater cost per bit.
  - => greater capacity, smaller cost per bit.
  - => greater capacity, greater access time.
- Desire=> greater capacity, faster access time, low cost per bit.
- Cannot be achieved with single memory component, solution is to organize computer memory into hierarchy.
- While moving down the hierarchy, we can observe the following:
  - => Increasing capacity.
  - => Decreasing cost per bit.
  - => Increasing access time / Decreasing speed.